WW-P-421

JULY 21, 1931

SUPERSEDING
F. S. No. 537
November 23, 1927

FEDERAL STANDARD STOCK CATALOGUE

Section IV

(Part 5)

FEDERAL SPECIFICATION

FOR

PIPE; WATER, CAST-IRON (BELL AND SPIGOT)

This specification was approved for promulgation by the Federal Specifications Board on July 21, 1931, for the use of the departments and independent establishments of the Government in the purchase of this commodity and shall become mandatory for all departments and independent establishments of the Government not later than March 1, 1932. It may be put into effect, however, at any earlier date after promulgation.

A. APPLICABLE FEDERAL SPECIFICATIONS.

A–1. The following Federal Specifications, or parts thereof that are below indicated, of the issue in effect on date of invitation for bids, shall form a part of this specification:

F. S. SS–C–191, Portland Cement.
F. S. QQ–M–151 (F. S. B. No. 339), General Specifications for Metals.

A–2. Any special requirements of the individual departments of the Government are noted under section H.

B. TYPES AND CLASSES.

B–1. This specification covers bell and spigot cast-iron water pipe in the following types:

Type I—Centrifugally cast in metal contact molds in 12-foot and 18-foot lengths.
Type II—Centrifugally cast in sand-lined molds in 16-foot, 16½-foot and 20-foot lengths.
Type III—Horizontally cast in green sand molds in 12-foot and 16-foot lengths.

B–2. This specification covers the following classes:

Class 150 for 150 pounds maximum water working pressure. (See par. I–6.)
Class 250 for 250 pounds maximum water working pressure. (See par. I–6.)

B–3. This specification covers sizes from 4 to 24 inches, inclusive, for types I and II, and from 4 to 12 inches, inclusive, for type III.

B–4. Pipe shall be furnished coated, unless cement lined or uncoated pipe is required in the invitation for bids.

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C. MATERIAL AND WORKMANSHIP.

C–1. All pipe shall be made of cast iron of good quality and of such character as shall make the pipe strong, tough, and of even grain and soft enough to admit of satisfactory drilling and cutting. Each pipe shall be smooth, free from cold shuts, scale, lumps, blisters, and sand holes and defects of every nature which unfit it for the use intended. It shall be straight and shall be true circles in section with its inner and outer surfaces concentric. No plugging, filling, "burning-in" or welding will be allowed.

D. GENERAL REQUIREMENTS.

D–1. *Lengths.*—Pipe shall have nominal laying lengths of 12 feet, 16 feet, 16½ feet, 18 feet, or 20 feet. Not more than 10 per cent of cut pipe shall be accepted. Such cut pipe shall be not more than 1 foot under nominal laying length.

D–2. *Cleaning.*—All pipe shall be thoroughly cleaned, and any rough spots in bells or on spigot ends shall be removed before coating or lining.

D–3. *Coating.*—Unless otherwise specified all pipe shall be completely coated inside and out with coal-tar pitch varnish, to which sufficient oil shall be added to make a smooth coating, tough and tenacious when cold, not "tacky" and not brittle.

D–4. *Cement mortar lining.*—When required in the contract or purchase order, pipe shall be lined with cement mortar. No coating shall be applied to the outside of pipe unless specifically provided for in the contract or order.

D–4a. *Preparation of pipe for lining.*—Pipe to be lined with cement mortar shall not be coated inside with tar or other asphaltum products. Its interior surface shall be thoroughly cleaned of all core sand, mud, grease, foreign materials, or any sharp projections of iron which might project through the lining. Pipe shall be tested hydrostatically before being lined.

D–4b. *Cement.*—The cement used for making cement mortar shall be standard Portland cement, complying in all respects with the requirements of Federal Specification SS–C–191 for Portland Cement, referred to in paragraph A–1.

D–4c. *Sand.*—The sand for mortar shall be clean, sharp, hard, and silicious sand, free from loam, clay, organic matter, or other foreign substance considered as deleterious for good mortar. The sand shall be well graded and when tested by laboratory sieves shall meet the following specifications:

Total passing 12-mesh sieve, 100 per cent.
Total passing 100-mesh sieve, not over 5 per cent.

D–4d. *Water.*—The water for tempering the cement mortar shall be free from harmful amounts of oil, acid, alkali, organic or vegetable matter.

D–4e. *Mortar.*—The cement mortar used for lining pipe shall be a mixture of the above-specified sand and cement in such proportions as to obtain a good, hard, dense lining, reasonably well bonded to the pipe, and with a smooth interior surface. (A mixture which has been found to give very satisfactory results consists of three

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parts cement to one part sand, by volume.) The cement mortar shall be thoroughly mixed, only sufficient water being added to form a workable mixture for placing in the pipe. Only sufficient cement mortar shall be mixed for the immediate requirements of lining.

D–4f. *Application of lining.*—Sufficient cement mortar shall be introduced to produce the required thickness of lining and shall be spread evenly over the interior surface of the pipe by any suitable means. A careful examination shall be made after this operation is completed to see that the inner surface of the pipe is completely covered with cement mortar. The shoulder of the bell and the end of the spigot may be covered with cement mortar by applying with a brush. Surplus cement mortar shall be removed from the interior of the bell so as not to interfere with proper keying of the joint. The work of lining the pipe shall be done in a building where the product shall be protected from the direct rays of the sun, and from extreme weather conditions, such as rain, frost, etc. The product shall not be put on the yard until the cement has set sufficiently to avoid injury or damage thereto. Patching of improperly lined pipe will not be permitted.

D–4g. *Smoothness of lining.*—The lining of pipe shall be smooth and substantially free from noticeable ridges, corrugations, projections, or depressions.

D–4h. *Thickness of lining.*—The thickness of lining shall be $\frac{1}{8}$ inch$+\frac{1}{8}, -0$ inch for sizes 12 inches and under, and $\frac{3}{16}$ inch$+\frac{1}{8}, -0$ inch for sizes over 12 inches. At the end of the pipe where the lining naturally tends to taper off to a thin edge, the full thickness of lining shall extend to within 1 inch of end of pipe. Linings of greater thickness shall be furnished when specified.

D–4i. *Curing of lining.*—Immediately after pipe is lined with cement mortar, it shall be protected in a suitable manner to prevent the too rapid withdrawal of moisture from the cement mortar and, if necessary, suitable means shall be provided to keep lining damp for a period of at least 24 hours after lining.

D–4j. No pipe shall be shipped until the lining is thoroughly set.

D–5. *Weighing.*—Each length of pipe shall be weighed and the weight plainly and indelibly marked on the pipe. Coated pipe shall be weighed after coating. Cement lined pipe shall be weighed before lining and the mark shall be placed on the outside surface.

D–6. *Hardness.*—The average Rockwell number shall not exceed 95 when tested in accordance with the method described in paragraph F–6.

D–7. *Handling.*—Care shall be taken in handling the pipes not to injure the coating, and no pipes or other material shall be placed in the pipes during transportation, or at any time after they have been coated.

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D–8. *Marking.*—In addition to the weight being marked on the pipe as previously specified, each pipe shall have distinctly cast or stamped into the metal on the face or outside surface of the bell or on body near the bell the manufacturer's mark, and the year in which the pipe was cast. When so specified in the invitation for bids each pipe shall also have cast upon it or stamped into the metal, a designated symbol.

E. DETAIL REQUIREMENTS.

E–1. *Tolerances in thickness.*—

E–1a. The tolerances in thickness of pipe, plus or minus, shall not exceed those listed below.

Nominal diameter in inches:	Tolerance, plus or minus (inch)
4	0.04
6	.045
8	.05
10	.055
12	.06
14, 16, 18, 20, 24	.08

E–1b. For all sizes of pipe, tolerances not exceeding 0.02 inch additional to above will be allowed for spaces not exceeding 8 inches in length in any direction.

E–2. *Tolerance in weight.*—The weight of no single pipe shall be less than the nominal tabulated weight by more than 5 per cent. The total weight of any order shall be not more than 2 per cent under nominal weight.

E–3. Type I—*Centrifugally cast in metal contact molds in 12-foot and 18-foot lengths.*—

E–3a. Type I pipe shall be of the bell and spigot type, centrifugally cast in metal contact molds with plain spigot end, and with lead groove and self-centering shoulder in the bell.

E–3b. *Annealing.*—Type I pipe, after withdrawing from machines, shall be annealed to meet the hardness limits specified in paragraph F–6.

E–3c. *Physical requirements.*—Type I pipe shall have a secant modulus of elasticity (see par. F–4b) not to exceed 15,000,000 pounds per square inch, with a corresponding modulus of rupture not less than 45,000 pounds per square inch for sizes 12 inches and under, and 40,000 pounds per square inch for sizes over 12 inches.

E–3d. *Tolerances in diameter.*—The inside diameters of the bells and the outside diameters of the spigot ends of Type I pipe shall not vary plus or minus from the tabulated dimensions by more than 0.06 inch for pipe 12 inches and less in nominal diameter and 0.08 inch for pipe 14 inches and larger.

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E–3e. *Dimensions and weights.*—Type I pipe shall conform to the dimensions and weights given in Figure 1 and Tables I and II for the respective classes and subject to the tolerances given in paragraphs E–1a, E–1b, E–2, and E–3d.

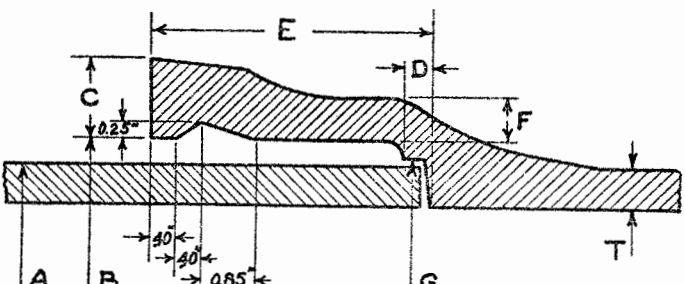


FIGURE 1.—Type I pipe, end details

TABLE I.—*Dimensions of Type I pipe (inches)*

Nominal inside diameter	Outside diameter of pipe	Inside diameter of bells	Total depth of bells	Depth of centering shoulder	Inside diameter of centering shoulder	Class 150, 150 pounds working pressure, 346-foot head			Class 250, 250 pounds working pressure, 576-foot head		
						Thickness of pipe	Thickness of bells		Thickness of pipe	Thickness of bells	
							At end	At shoulder		At end	At shoulder
	A	B	E	D	G	T	C	F	T	C	F
4	4.80	5.60	3.30	0.30	4.93	0.34	1.06	0.48	0.38	1.06	0.48
6	6.90	7.70	3.88	.38	7.13	.37	1.13	.52	.43	1.13	.52
8	9.05	9.85	4.38	.38	9.31	.42	1.18	.57	.50	1.18	.57
10	11.10	11.90	4.38	.38	11.43	.47	1.23	.63	.57	1.23	.63
12	13.20	14.00	4.38	.38	13.43	.50	1.28	.69	.62	1.28	.69
14	15.65	16.45	4.50	.50	15.85	.55	1.35	.73	.69	1.56	.93
16	17.80	18.80	4.50	.50	18.00	.60	1.43	.77	.75	1.63	1.00
18	19.92	20.92	4.50	.50	20.12	.65	1.53	.82	.83	1.68	1.09
20	22.06	23.06	4.50	.50	22.26	.65	1.63	.86	.88	1.78	1.18
24	26.32	27.32	4.50	.50	26.53	.76	1.83	1.03	1.00	2.00	1.38

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TABLE II.—*Nominal weights of Type I pipe (pounds)*

Nominal inside diameter	Class 150, 150 pounds working pressure, 346-foot head				Class 250, 250 pounds working pressure, 576-foot head			
	12-foot nominal laying length		18-foot nominal laying length		12-foot nominal laying length		18-foot nominal laying length	
	Weight of pipe	Weight per foot with bell	Weight of pipe	Weight per foot with bell	Weight of pipe	Weight per foot with bell	Weight of pipe	Weight per foot with bell
4	195	16.4	285	15.9	220	18.4	325	17.9
6	315	26.3	460	25.5	350	29.3	515	28.5
8	475	39.4	690	38.3	545	45.5	800	44.3
10	640	53.3	935	51.8	760	63.3	1,115	61.9
12	810	67.4	1,180	65.6	990	82.5	1,450	80.7
14	1,060	88.5	1,555	86.3	1,320	110.0	1,930	107.3
16	1,320	110.2	1,935	107.4	1,635	136.1	2,390	132.7
18	1,595	132.8	2,330	129.6	2,015	168.0	2,950	163.9
20	1,860	155.0	2,720	151.0	2,365	197.1	3,465	192.4
24	2,480	206.8	3,630	201.6	3,200	266.8	4,690	260.6

NOTE.—Calculated weight of pipe has been rounded off to nearest 5 pounds. Weight per foot is based on weight of pipe computed to nearest pound.

E–4. Type II.—*Centrifugally cast in sand-lined molds in 16-foot 16½-foot, and 20-foot lengths.*—

E–4a. Type II pipe shall be of the bell and spigot type, centrifugally cast in sand-lined molds; with bead cast on spigot end; and with lead groove in the bell.

E–4b. *Physical requirements.*—Type II pipe shall have a secant modulus of elasticity (see par. F–4b) not to exceed 10,000,000 pounds per square inch, with a corresponding modulus of rupture not less than 40,000 pounds per square inch.

E–4c. *Tolerances in diameter.*—The inside diameters of the bells and the outside diameters of the spigot ends (exclusive of bead) of type II pipe shall not vary plus or minus from the tabulated dimensions by more than 0.06 inch for pipe 12 inches and less in nominal, diameter, and 0.08 inch for pipe 14 inches and larger.

E–4d. *Dimensions and weights.*—Type II pipe shall conform to the dimensions and weights given in Figure 2 and Tables III and IV for the respective classes and subject to the tolerances given in para-graphs E–1a, E–1b, E–2, and E–4c.

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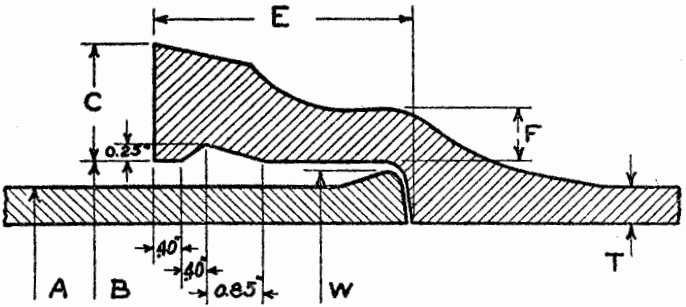


FIGURE 2.—Type II pipe, end details

TABLE III.—*Dimensions of Type II pipe (inches)*

Nominal Inside diameter	Outside diameter of pipe	Outside diameter of spigot end, maximum	Inside diameter of bells	Depth of bells	Class 150, 150 pounds working pressure, 346-foot head			Class 250, 250 pounds working pressure, 576-foot head		
					Thickness of pipe	Thickness of bells		Thickness of pipe	Thickness of bells	
						At end, minimum	At shoulder minimum		At end, minimum	At shoulder minimum
	A	W	B	E	T	C	F	T	C	F
4	4.80	5.36	5.60	3.50	0.34	1.28	0.44	0.38	1.28	0.44
6	6.90	7.46	7.70	3.50	.37	1.39	.48	.43	1.39	.48
8	9.05	9.61	9.85	4.00	.42	1.47	.52	.50	1.47	.62
10	11.10	11.66	11.90	4.00	.47	1.59	.57	.57	1.59	.69
12	13.20	13.76	14.00	4.00	.50	1.75	.62	.62	1.75	.74
14	15.65	16.23	16.45	4.00	.55	1.47	.62	.69	1.62	.77
16	17.80	18.38	18.80	4.00	.60	1.52	.67	.75	1.72	.82
18	19.92	20.50	20.92	4.00	.65	1.62	.72	.83	1.82	.87
20	22.06	22.64	23.06	4.00	.68	1.77	.77	.88	1.92	.92
24	26.32	26.90	27.32	4.00	.76	1.92	.82	1.00	2.07	.97

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TABLE IV.—*Nominal weights of Type II pipe (pounds)*

Nominal inside diameter	Class 150, 150 pounds working pressure, 346-foot head						Class 250, 250 pounds working pressure, 576-foot head					
	16-foot nominal laying length		16½-foot nominal laying length		20-foot nominal laying length		16-foot nominal laying length		16½-foot nominal laying length		20-foot nominal laying length	
	Weight of pipe	Weight per foot with bell	Weight of pipe	Weight per foot with bell	Weight of pipe	Weight per foot with bell	Weight of pipe	Weight per foot with bell	Weight of pipe	Weight per foot with bell	Weight of pipe	Weight per foot with bell
4	255	16.1	265	16.0	------	------	290	18.1	300	18.0	------	------
6	410	25.7	425	25.6	------	------	460	28.7	475	28.6	------	------
8	615	38.6	635	38.5	------	------	715	44.6	735	44.5	------	------
10	835	52.2	860	52.1	------	------	995	62.3	1,025	62.2	------	------
12	1,055	66.1	1,090	65.9	------	------	1,300	81.1	1,335	81.0	------	------
14	1,390	86.9	1,430	86.7	------	------	1,730	108.0	1,780	107.8	------	------
16	1,730	108.1	1,780	107.9	2,140	106.9	2,135	133.6	2,200	133.3	2,640	132.1
18	2,085	130.4	2,150	130.2	2,580	128.9	2,640	164.9	2,715	164.7	3,265	163.2
20	2,430	152.0	2,505	151.7	3,005	150.2	3,095	193.6	3,190	193.2	3,830	191.5
24	3,245	202.9	3,340	202.5	4,010	200.5	4,195	262.1	4,320	261.7	5,185	259.3

NOTE.—Calculated weight of pipe has been rounded off to nearest 5 pounds. Weight per foot is based on weight of pipe computed to nearest pound.

E–5. Type III.—*Horizontally cast in green-sand molds in 12-foot and 16-foot lengths.*—

E–5a. Type III pipe shall be of the bell and spigot type, horizontally cast in green-sand molds with green-sand cores and multiple gates, and with spigot end bead and lead groove in the bell.

E–5b. *Physical requirements.*—Type III pipe shall have a secant modulus of elasticity (see paragraph F–4b) not to exceed 12,500,000 pounds per square inch, with a corresponding modulus of rupture not less than 42,500 pounds per square inch.

E–5c. *Tolerances in diameter.*—The inside diameters of bells and the outside diameters of the spigot ends (exclusive of bead) of Type III pipe shall not vary plus or minus from the tabulated dimensions by more than 0.06 inch for pipe 12 inches and less in nominal diameter.

E–5d. *Dimensions and weights.*—Type III pipe shall conform to the dimensions and weights given in Figure 3 and Tables V and VI for the respective classes and subject to the tolerances given in paragraphs E–1a, E–1b, E–2, and E–5c.

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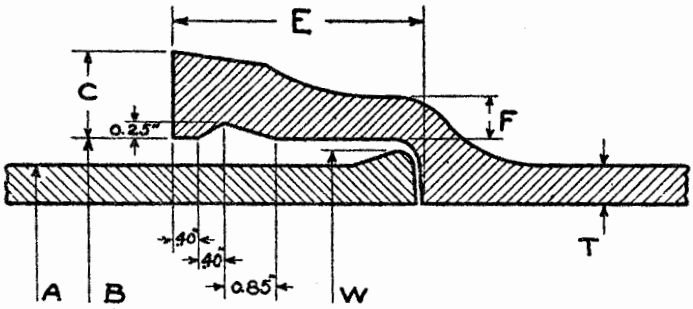


FIGURE 3.—Type III pipe, end details

TABLE V.—*Dimensions of Type III pipe (Inches)*

Nominal inside diameter	Outside diameter of pipe	Outside diameter of spigot end, Maximum	Inside diameter of bell	Thickness of pipe		Depth of bell	Thickness of bell at end	Thickness of bell at shoulder
				Class 150	Class 250			
	A	W	B	T	T	E	C	F
4	4.80	5.24	5.60	0.34	0.38	3.50	0.88	0.55
6	6.90	7.34	7.70	.37	.43	3.50	1.00	.60
8	9.05	9.61	9.85	.42	.50	4.00	1.13	.65
10	11.10	11.66	11.90	.47	.57	4.00	1.25	.70
12	13.20	13.76	14.00	.50	.62	4.00	1.38	.75

TABLE VI.—*Nominal weights of Type III pipe (Pounds)*

Nominal inside diameter	Class 150, 150 pounds working pressure, 346-foot head				Class 250, 250 pounds working pressure, 576-foot head			
	12-foot nominal laying length		16-foot nominal laying length		12-foot nominal laying length		16-foot nominal laying length	
	Weight of pipe	Weight per foot with bell	Weight of pipe	Weight per foot with bell	Weight of pipe	Weight per foot with bell	Weight of pipe	Weight per foot with bell
4	195	16.4	255	16.1	220	18.4	290	18.1
6	315	26.3	410	25.7	350	29.3	460	28.7
8	475	39.4	615	38.6	545	45.5	715	44.6
10			835	52.2			995	62.3
12			1,055	66.1			1,300	81.1

NOTE.—Calculated weight of pipe has been rounded off to nearest 5 pounds. Weight per foot is based on weight of pipe computed to nearest pound.

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F. METHOD OF SAMPLING, INSPECTION, AND TESTS.

F–1. Any requirements applicable to the individual departments are noted under section H.

F–2. *Casting record.*—A record of the melting and pouring temperatures of the iron shall be furnished when requested.

F–3. Chemical analysis shall be made by the manufacturer from each heat to determine graphitic and combined carbon or total carbon if the separate determinations are not available, manganese, phosphorus, sulphur, and silicon, and duplicate copies of test reports shall be furnished when requested. Sulphur shall not exceed 0.10 per cent and phosphorus 0.90 per cent in either ladle or inspection analysis.

F–4. *Tests of material.*—

F–4a. From each 300 lengths of pipe, or fraction thereof, of each size in the contract or order, one length of pipe shall be selected by the inspector before coating. From each sample pipe there shall be cut and machined one test strip 12 inches long, 0.50 inch deep, and the full thickness of the shell in width. This shall be tested as a beam (with machined surfaces top and bottom) on supports 10 inches apart with load applied at two points 3⅓ inches from the supports. The strip shall be accurately calipered at point of rupture and stress calculated by the formula

$$S=\frac{PLc}{6I} \text{ or for the above specimen } S=\frac{40P}{b}$$

F–4b. The secant modulus of elasticity at the breaking load shall be calculated by the formula

$$E=\frac{23PL^3}{1{,}296Iy} \text{ or for the above specimen } E=S\frac{42.6}{y}$$

In these formulas—

S=modulus of rupture.
E=modulus of elasticity.
P=total load.
L=length of span.
c=distance to extreme fiber.
I=moment of inertia.
b=width of specimen (thickness of pipe).
d=depth of specimen.
y=center deflection at load P.

F–5. *Hydrostatic tests.*—Each length of pipe shall be subjected to a hydrostatic pressure of 500 pounds per square inch. The pipe shall be under this pressure at least one-half minute, and while under pressure shall be subjected to a hammer test. Any pipes showing defects by leaking, sweating, or otherwise, shall be rejected.

F–6. *Rockwell tests.*—Upon the machined edges of each test strip there shall be made at well-distributed points not less than three determinations, using a 1.59-millimeter (1/16-inch) ball and 100 kilogram load. The average Rockwell number shall not exceed 95. An additional determination shall also be made for pipe cast in metal contact molds, upon the outside of each pipe, using a portable type machine. The equivalent Rockwell number for the exterior shall not exceed 95.

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F–7. *Cement lining.*—Not less than 1 per cent of pipe shall be tested, and in the event defects are discovered in the lining, testing shall proceed until the lining complies with the specification.

F–7a. The thickness of lining may be determined by means of spear measurements, using a hardened steel point not greater than one-sixteenth inch in diameter. The inspector shall pierce the lining immediately after it is placed in the pipe, and before cement has set, at four diametrically opposite points of the pipe at bell and spigot ends, making two sets of measurements at each end. The first set shall be not greater than 4 inches from the respective ends of the pipe and the second set shall be made as far into the interior of the pipe as can readily be obtained by reaching into the pipe without injuring the lining. All measurements shall be within the limits specified.

F–7b. For lining of the above specified thickness or of greater thickness, failure of the lining to completely adhere to the wall of the pipe shall not be cause for rejection, if the lining conforms to these specifications in all other respects.

F–8. If any test specimen shows defective machining or obvious lack of continuity of metal, it may be discarded and replaced by another specimen selected by the inspector. If any of the test specimens selected fail to meet the requirements specified and the purchaser permits a retest, at least twice the number of specimens that failed shall be selected by the inspector for replacement, all of which shall meet the requirements.

G. PACKING AND MARKING OF SHIPMENTS.

G–1. Any special requirements of the individual departments are noted under section H.

G–2. *Packing.*—Unless otherwise specified, the subject commodity shall be prepared for standard commercial shipment so as to insure acceptance by common or other carriers, for safe transportation, at the lowest rate, to the point of delivery.

G–3. *Marking.*—Unless otherwise specified, shipments shall be marked with the number of the contract or order.

H. REQUIREMENTS APPLICABLE TO INDIVIDUAL DEPARTMENTS.

H–1. The following departmental specifications of the issue in effect on date of invitation for bids shall respectively form a part of this specification:

H–1a. Army: United States Army Specification No. 100–2, Standard Specification for Marking Shipments.

H–1b. Navy: General Specification for Inspection of Materials, together with Appendix II (Metals), copies of which may be obtained without cost upon application to the Bureau of Supplies and Accounts, Navy Department, Washington, D. C. Appendix II (Metals) is applicable for Navy Department purchases, in lieu of Federal Specification QQ–M–151 (F. S. B. No. 339), referred to in paragraph A–1 of this specification.

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I. NOTES.

I–1. Invitations for bids should state the number of linear feet of pipe desired and payment should be made on a per foot basis. No bids should be accepted on a tonnage basis.

I–2. Award of contract ordinarily will be made on the basis of *evaluated* cost per foot, in order to take into account the cost of jointing materials required.

I–2a. The cost per foot may be evaluated from the price bid per foot by the following formula:

$$E=A+KP$$

in which—

E=evaluated cost per foot in dollars.
A=bid price per foot in dollars.
K=constant found in Table VII opposite diameter of pipe and under length of pipe bid upon.
P=The price of lead [1] in *cents* per pound current upon date of invitation for bids. The current price of lead may be obtained from the publication "Iron Age". Only New York prices shall be considered.

TABLE VII.—*Values of constant "K" for evaluating prices bid per foot of cast-iron water pipe*

[Based on total weight of lead and jute at 1 cent per pound required for one joint]

Nominal pipe diameter (inches)	Length of pipe in feet				
	12	16	16½	18	20
4	0.0064	0.0048	0.0047	0.0043	0.0038
6	.0088	.0066	.0064	.0059	.0053
8	.0114	.0086	.0083	.0076	.0068
10	.0138	.0103	.0101	.0092	.0083
12	.0163	.0123	.0120	.0109	.0098
14	.0190	.0143	.0139	.0127	.0114
16	.0258	.0193	.0189	.0172	.0155
18	.0290	.0218	.0212	.0193	.0174
20	.0319	.0239	.0234	.0212	.0191
24	.0379	.0284	.0278	.0252	.0227

Example.—If the bid price on 12-inch diameter pipe in 18-foot lengths is $1.25 per foot, and the current price of lead is 6½ cents per pound, the evaluated price per foot of pipe

$E=\$1.25+\$(.0109\times 6½)=\$1.25+0.0708=\$1.3208$ per foot.

I–3. If special symbol is desired (see par. D–8), it should be so stated or indicated in the invitation for bids. If not so stated, it will be assumed that none is desired.

I–4. For lines for water known to cause serious tuberculation, it is recommended that pipe be not coated, and that it be lined with cement mortar.

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[1] The price of jute need not be considered.

I–4a. When linings other than those specified in section D–4 are required, it should be so stated in the invitation for bids.

I–5. When American Water Works Association pipe is specifically required by the requisition or order, such pipe shall conform to the requirements and tests of the A. W. W. A. specification. In such cases the desired A. W. W. A. class should be stated.

I–6. This specification is designed for average waterworks construction conditions. Where construction conditions involving excessive vibration, shock, bad trench conditions, corrosion, etc., are contemplated, due consideration should be given to the use of pipe of heavier weight.

I–6a. Cast-iron pipe of lighter or heavier weight than specified herein or with other than bell and spigot joint should be purchased under a separate specification.

I–7. As far as known at the time of promulgation of this specification Type III pipe was produced regularly by but one company.

I–8. Bids shall clearly indicate the type and length of pipe which bidders will furnish.

I–9. It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

I–10. This specification governs all United States Government purchases of this commodity. Users are requested to retain this specification in permanent file until a revision is issued.

I–11. Interested parties are advised that an Alphabetical Index of Federal Specifications may be obtained upon application, as noted in paragraph next below, price to be obtained from the Superintendent of Documents.

I–12 Copies of this specification, together with Federal Specification SS–C–191 referred to in paragraph A–1, may be obtained upon application, accompanied by money order, or coupon, or cash, to Superintendent of Documents, Government Printing Office, Washington, D. C. Price of this specification 5 cents. SS–C–191, 10 cents.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility or any obligation whatsoever, and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specification, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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